Probabilistic Graphical Models I: Bayesian Network Representation

Statistical Data Mining II

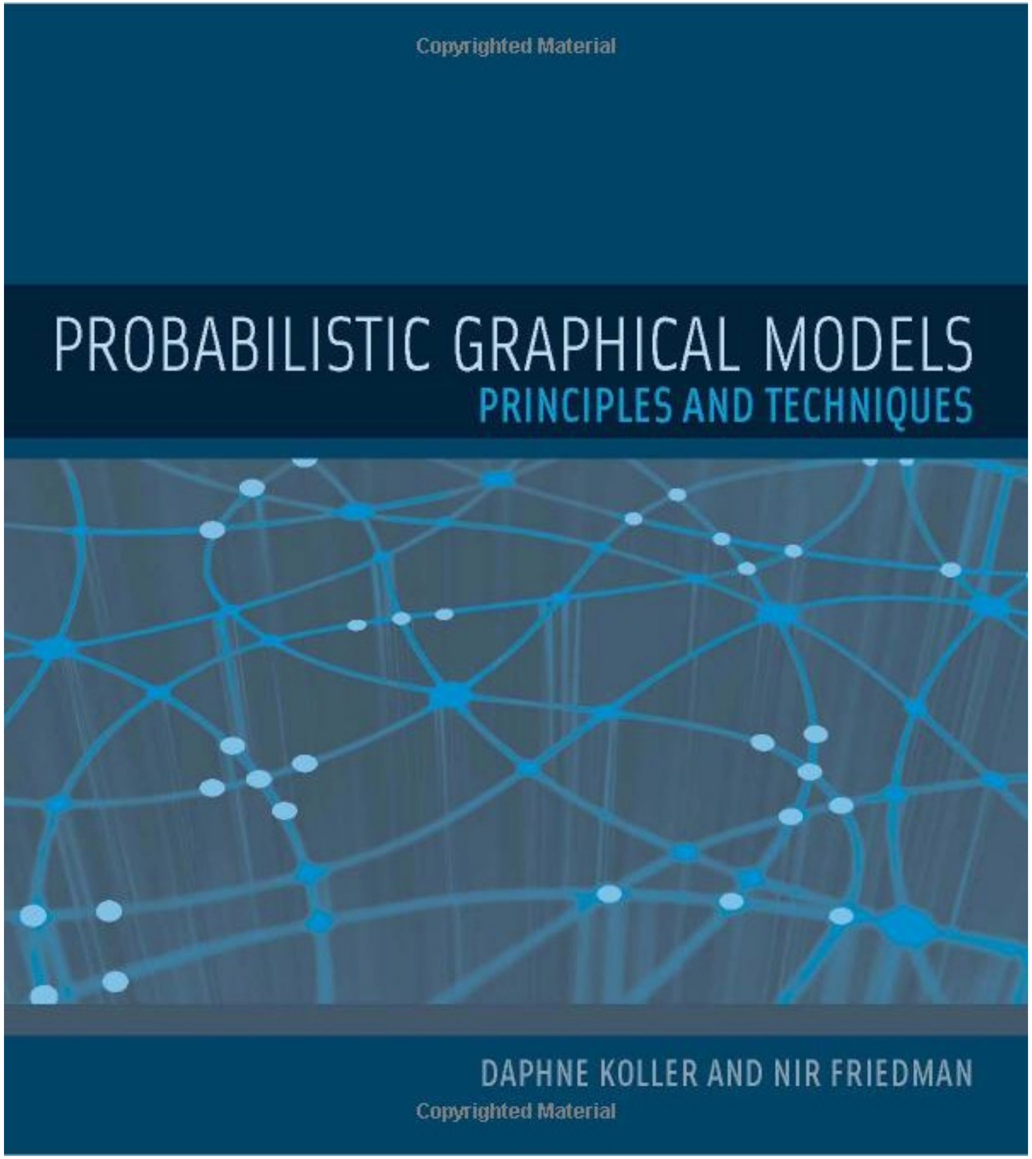
Rachael Hageman Blair

We are re-entering the world of prediction.

Probabilistic Graphical Models (PGM):

- Infer the topology of associations
- Can be used to make real-valued predictions
- And predictions regarding associations

Flexible modeling paradigm, a cornerstone in systems biology.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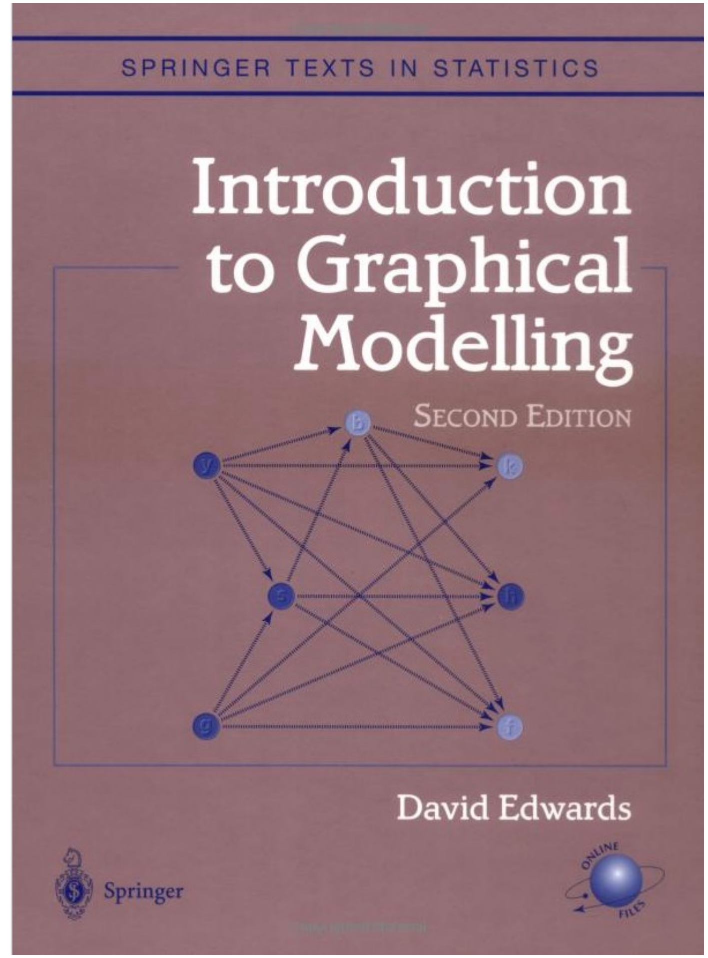
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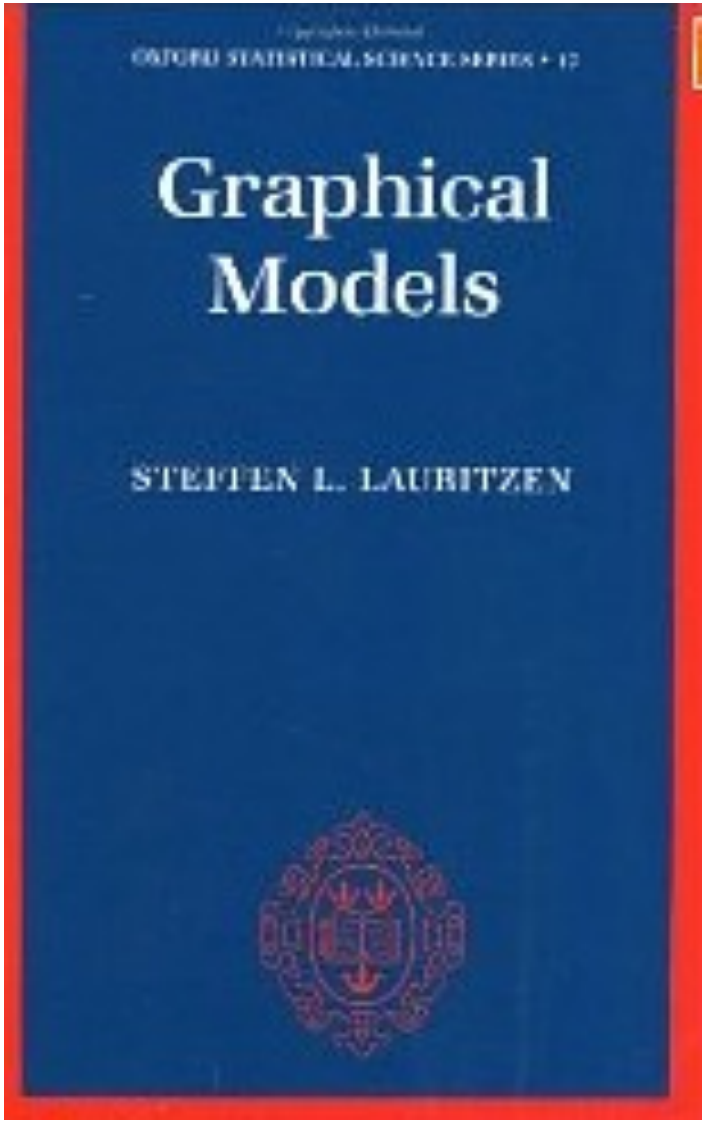
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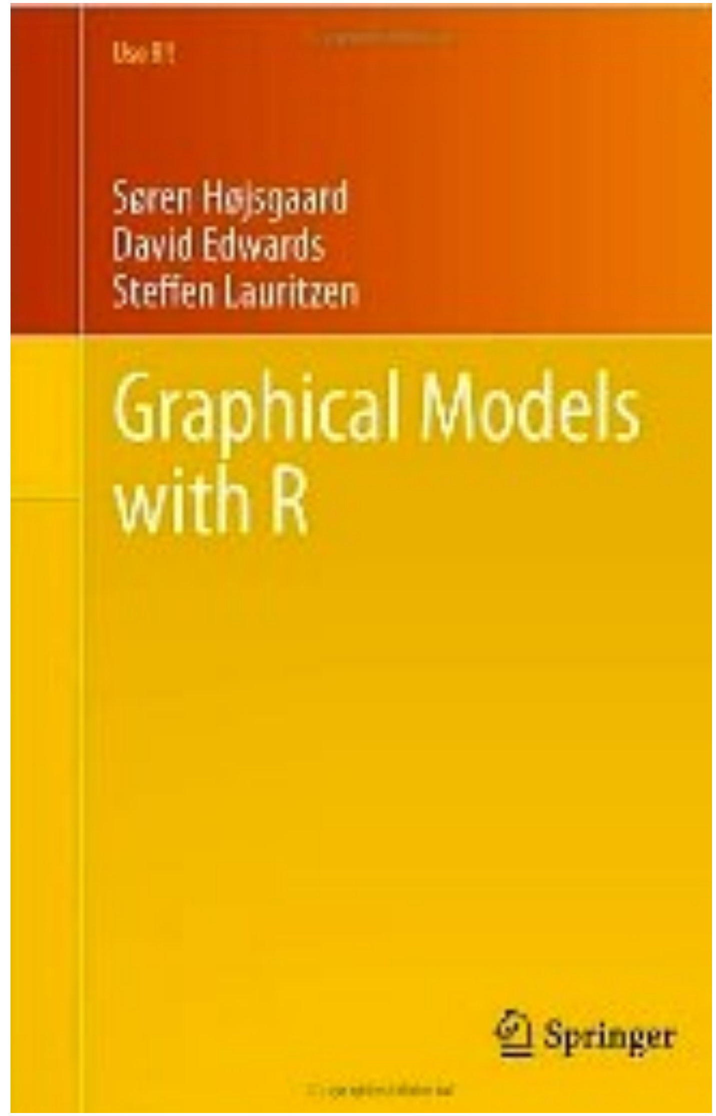
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Outline

- Motivation
- Preliminaries: Properties
- Preliminaries: Working with a distribution
- Preliminaries: Factors
- Bayesian Networks
- Reasoning
- Flow of Probabilistic Reasoning

Motivation

Millions of pixels

Family History (Genetics)
Predisposing Factors
Test Results

Each needs to be labeled
{grass, sky, water, cow, horse, … }

Infer: diagnosis, and treatment outcomes

Common Threads: Large number of variables, uncertainty.

Motivation

Uncertainty:

- **Partial knowledge of the state of the world.**
- **Partial data available.**
- **Phenomena not covered by the model.**
- **Noisy observations.**
- **Modeling limitations of complicated systems.**

Probabilistic Graphical Models:

- **Declarative representation with clear semantics.**
- **Powerful reasoning patterns.**

Motivation

Random Variables pertaining to a course:

- **Difficulty**
- **Intelligence**
- **Grade**
- **Letter of Recommendation**
- **SAT**

Motivation

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- **Difficulty**
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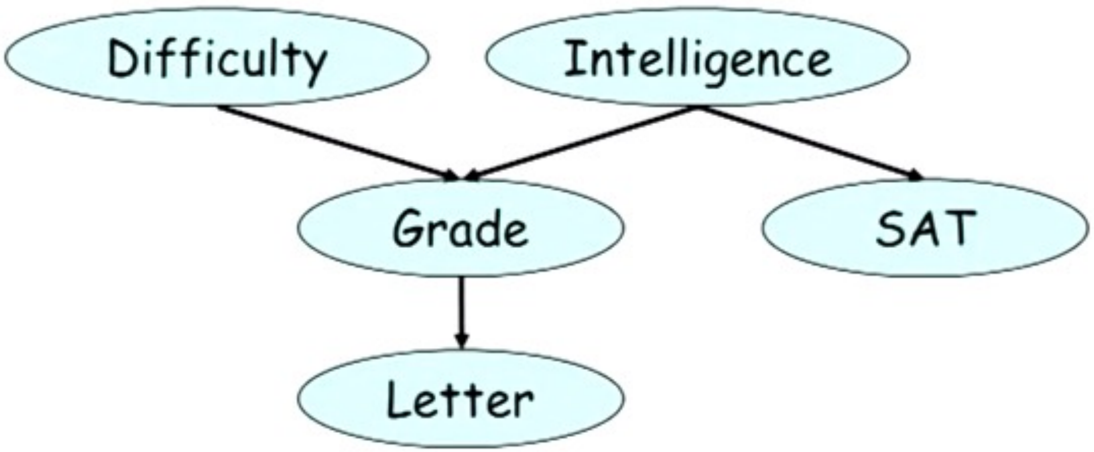


Q: How do we encode this into a probability distribution?

Motivation

Complex Systems:

- Modeled with Random Variables
- **Goal:** obtain the relationship between a set of random variables, $X_1, X_2, \ldots, X_N$, in terms of a a joint probability distribution

$$P\left(X_1, X_2, \ldots, X_N\right),$$

which will capture relationships and uncertainty between variables.

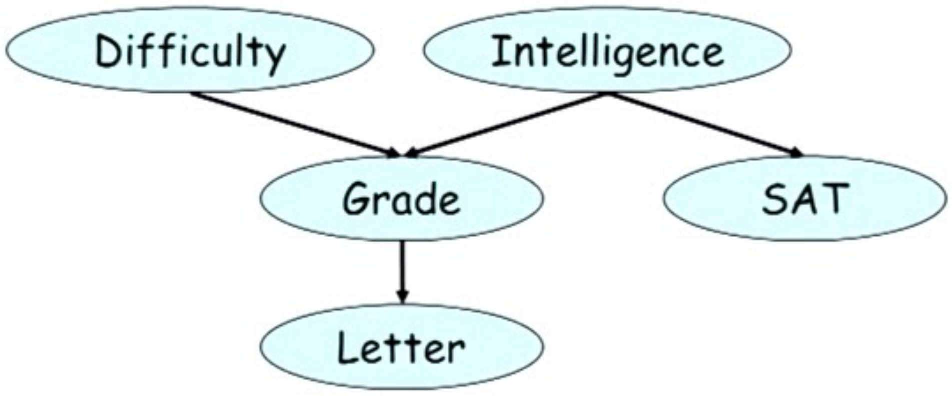
*** Note: even in the simplest case, when all variables are binary, there are 2^N possible states. Things get big fast.

Will have to exploit structural assumptions on the relationships in order to make the problem manageable.

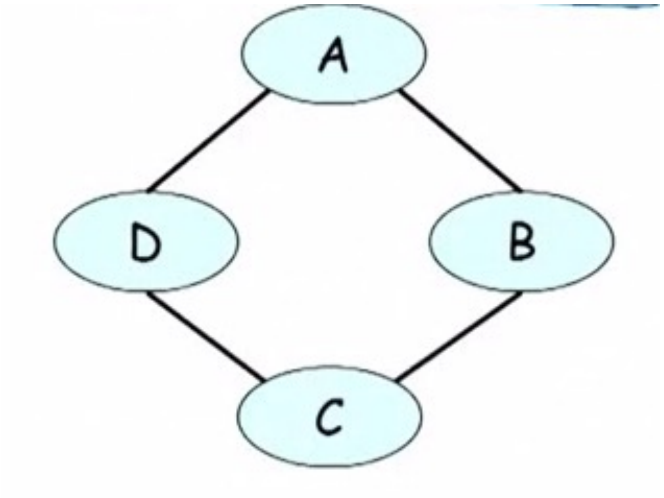
Motivation

Graphical Models:

Bayesian Networks

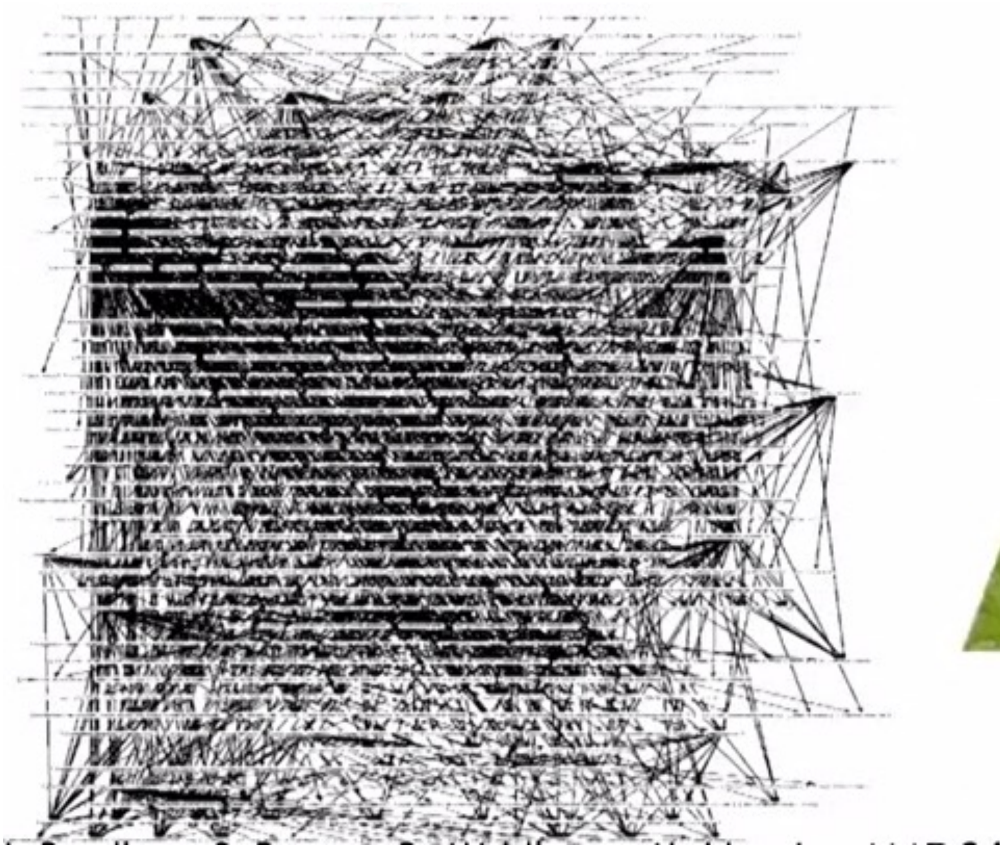


Markov Networks

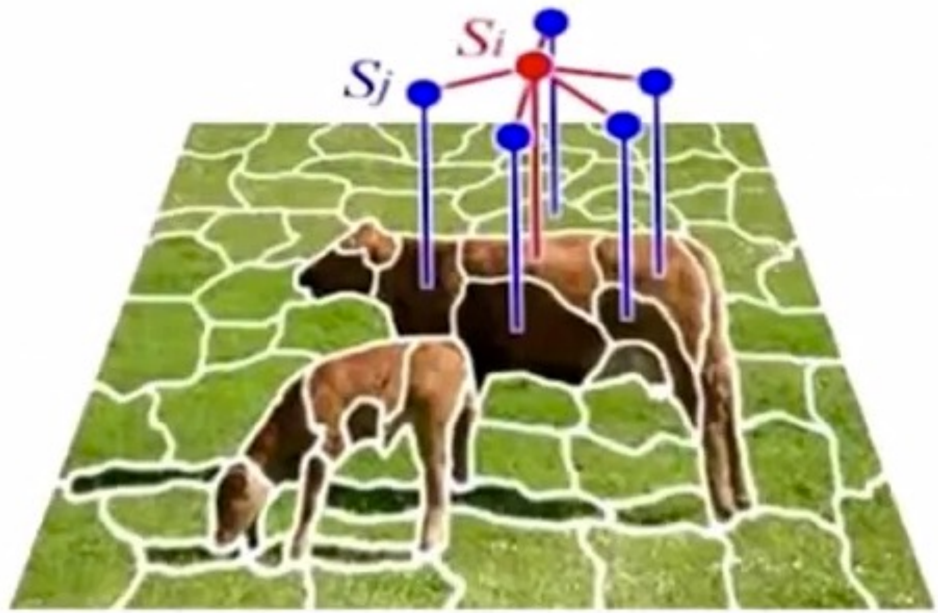


Motivation

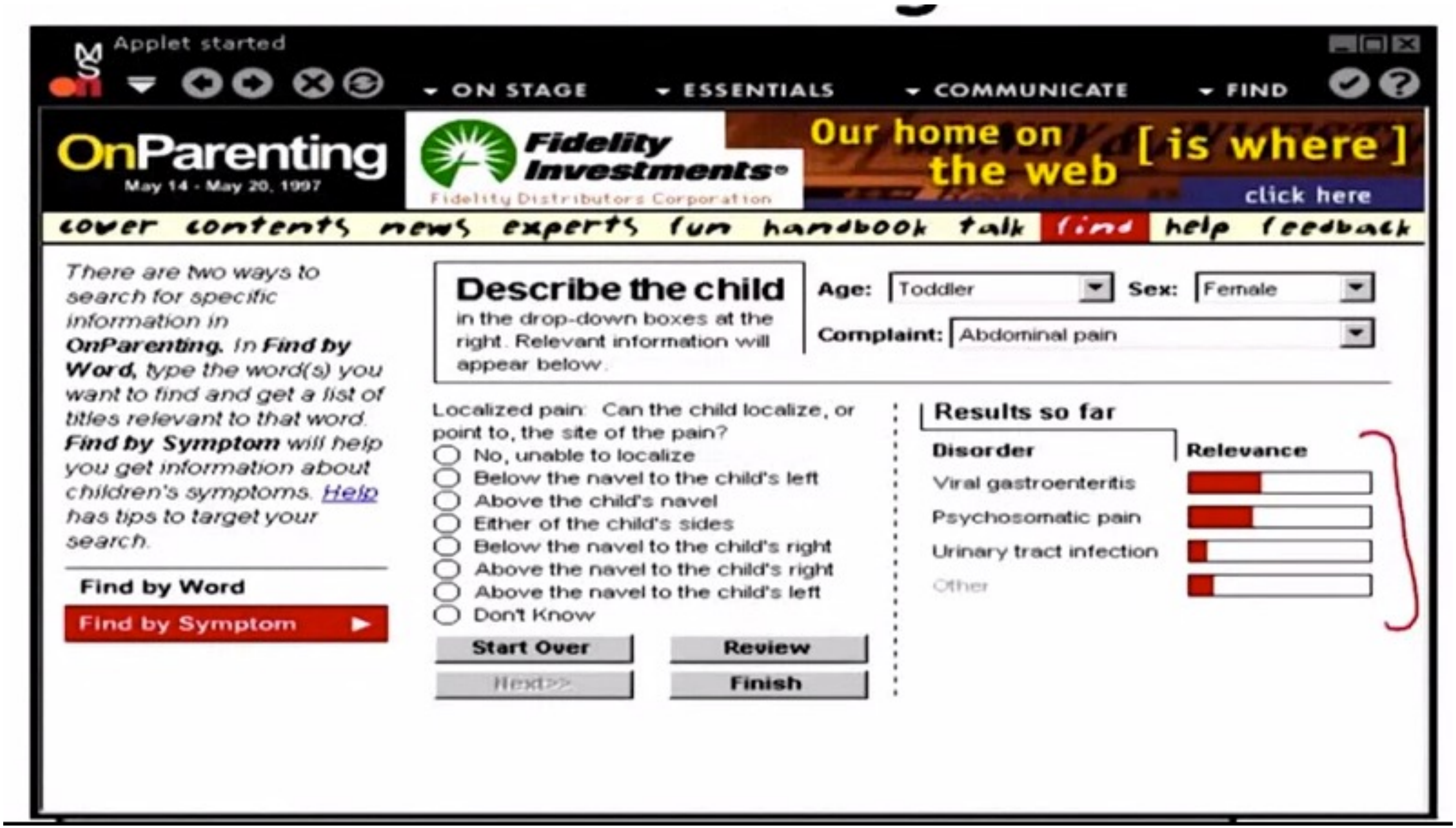
Bayesian Networks
900+ edges,
CPCS: medical diagnoses of diseases



Markov Networks
Image segmentation
Rv represent labels, super-pixels



Motivation



Applet started
ON STAGE
ESSENTIALS
COMMUNICATE
FIND
OnParenting
May 14 - May 20, 1997
Fidelity Investments
Fidelity Distributors Corporation
Our home on the web
[is where]
click here
cover contents news experts fun handbook talk find help feedback
There are two ways to search for specific information in OnParenting. In Find by Word, type the word(s) you want to find and get a list of titles relevant to that word. Find by Symptom will help you get information about children's symptoms. Help has tips to target your search.
Find by Word
Find by Symptom
Describe the child
in the drop-down boxes at the right. Relevant information will appear below.
Age: Toddler
Sex: Female
Complaint: Abdominal pain
Localized pain: Can the child localize, or point to, the site of the pain?
No, unable to localize
Below the navel to the child's left
Above the child's navel
Either of the child's sides
Below the navel to the child's right
Above the navel to the child's right
Above the navel to the child's left
Don't Know
Start Over
Review
Next>>
Finish
Results so far
Disorder
Relevance
Viral gastroenteritis
Psychosomatic pain
Urinary tract infection
Other

Motivation

Many Applications:

- Medical Diagnosis
- Fault Diagnosis
- Natural processing language
- Traffic Analysis
- Social network models
- Computer vision e.g., image segmentation, 3D reconstruction
- Robotics

Preliminaries: Properties

Conditional Probability Motivation

- **Example:** Suppose we want to reason about a students intelligence and their final grade. We can define the event

  α: All students with a grade of A.

  β: All students with high intelligence.

  Using distributions, we can reason about these two events, and their intersection, $\alpha \cap \beta$, (intelligent students that got an A), but this tells us nothing about how to "update out beliefs based on new evidence".

  "A student receives an A, what can that tell us about her intelligence"?

Preliminaries: Properties

Conditional Probability Motivation:

More precisely, when we learn about an event that is TRUE, how does that change the probability of occurring? The answer is through the notion of conditional probability.

- The conditional probability of α given β is as follows:

$$P(\alpha \mid \beta) = \frac{P(\alpha \cap \beta)}{P(\beta)}$$

which is defined for $P(\beta) > 0$.

Preliminaries: Properties

Bayes Rule:

It follows directly that:

$$P(\alpha \mid \beta) = \frac{P(\beta \mid \alpha) P(\alpha)}{P(\beta)}$$

which is defined for $P(\beta) > 0.$

*Many classic examples: disease screening

Preliminaries: Properties

Independence

- We say an event α is independent of event β iff $P(\alpha \mid \beta) = P(\alpha)$ or $P(\beta) = 0$.

  Furthermore, $\alpha \perp \beta$, iff $P(\alpha\beta) = P(\alpha)P(\beta)$.

*Independence is useful, however, uncommon. It is not very common to come across two independent events.

More common: two events are independent, given a third event. This phenomena is known as "conditional independence".

Preliminaries: Properties

More common: two events are independent, given a third event. This phenomena is known as "conditional independence".

- A student applies to grad school at MIT and Harvard. We want to reason about acceptance. Define "MIT" as admitted to MIT and "Harvard" as admitted to Harvard.

  - These events are not independent. If we learn the student got into MIT, then she has a better shot (probability) of getting into Harvard because she must be good.

  - Now suppose they both base their decisions ONLY on grades.

$$P\left(\mathrm{MIT} \mid \mathrm{Harvard}, \mathrm{GradeA}\right) = P(\mathrm{MIT} \mid \mathrm{GradeA})$$

MIT acceptance is **conditionally independent** of Harvard, given the grade A.

Preliminaries: Properties

Conditional Independence

- We say an event α is conditionally independent of event β given an event γ, denoted $(\alpha \perp \beta \mid \gamma)$, if

$$P(\alpha \mid \beta \cap \gamma) = P(\alpha \mid \gamma)$$

or if $P(\beta \cap \gamma) = 0.$

- The relation $\alpha \perp \beta \mid \gamma$ is satisfied if and only if:

$$P\left(\alpha \cap \beta \mid \gamma\right) = P\left(\alpha \mid \gamma\right) P\left(\beta \mid \gamma\right).$$

Preliminaries: Working with Distributions

A closer look at the joint distribution:

2 Possible Values

- Intelligence (I)
  - i^0 (low), i^1 (high),
- Difficulty (D)
  - d^0 (easy), d^1 (hard)
- Grade (G)
  - g^1 (A), g^2 (B), g^3 (C)

2 Possible Values

3 Possible Values

Preliminaries: Working with Distributions

A closer look at the joint distribution P(I,D,G):

- Intelligence (I)
  - i^0 (low), i^1 (high),
- Difficulty (D)
  - d^0 (easy), d^1 (hard)
- Grade (G)
  - g^1 (A), g^2 (B), g^3 (C)

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^0	g^2	0.168
i^0	d^0	g^3	0.126
i^0	d^1	g^1	0.009
i^0	d^1	g^2	0.045
i^0	d^1	g^3	0.126
i^1	d^0	g^1	0.252
i^1	d^0	g^2	0.0224
i^1	d^0	g^3	0.0056
i^1	d^1	g^1	0.06
i^1	d^1	g^2	0.036
i^1	d^1	g^3	0.024

2 x 2 x 3 = 12 possible probabilities
12 parameters
11 independent parameters

Must sum to one

Preliminaries: Working with Distributions

A closer look at the joint distribution P(I,D,G):

Lets assume a student got an A.
We can “condition” of g1.

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^0	g^2	0.168
i^0	d^0	g^3	0.126
i^0	d^1	g^1	0.009
i^0	d^1	g^2	0.045
i^0	d^1	g^3	0.126
i^1	d^0	g^1	0.252
i^1	d^0	g^2	0.0224
i^1	d^0	g^3	0.0056
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i^1	d^1	g^3	0.024

Preliminaries: Working with Distributions

A closer look at the joint distribution:

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I	D	G	Prob.
i^0	d^0	g^1	0.126
~~i^0~~	~~d^0~~	~~g^2~~	~~0.168~~
~~i^0~~	~~d^0~~	~~g^3~~	~~0.126~~
i^0	d^1	g^1	0.009
~~i^0~~	~~d^1~~	~~g^2~~	~~0.045~~
~~i^0~~	~~d^1~~	~~g^3~~	~~0.126~~
i^1	d^0	g^1	0.252
~~i^1~~	~~d^0~~	~~g^2~~	~~0.0224~~
~~i^1~~	~~d^0~~	~~g^3~~	~~0.0056~~
i^1	d^1	g^1	0.06
~~i^1~~	~~d^1~~	~~g^2~~	~~0.036~~
~~i^1~~	~~d^1~~	~~g^3~~	~~0.024~~

Preliminaries: Working with Distributions

Conditioning:

Lets assume a student got an A.
We can “condition” of g1.

Reduction!
No longer a distribution as we have written it.

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^1	g^1	0.009
i^1	d^0	g^1	0.252
i^1	d^1	g^1	0.06

Preliminaries: Working with Distributions

Conditioning:

I	D	G	Prob.
i^0	d^0	g^1	0.126
i^0	d^1	g^1	0.009
i^1	d^0	g^1	0.252
i^1	d^1	g^1	0.06

$P(I, D, g^1)$

$$\sum = 0.447$$

Un-normalized measure

Preliminaries: Working with Distributions

Conditioning:

I	D	G	Prob.	
i^0	d^0	g^1	0.126	$/ \ 0.447$
i^0	d^1	g^1	0.009	$/ \ 0.447$
i^1	d^0	g^1	0.252	$/ \ 0.447$
i^1	d^1	g^1	0.06	$/ \ 0.447$

$P(I, D, g^1)$

$$\sum = 0.447$$

I	D	Prob.
i^0	d^0	0.282
i^0	d^1	0.02
i^1	d^0	0.564
i^1	d^1	0.134

$P(I, D \mid g^1)$

Preliminaries: Working with Distributions

Marginalization :

Marginalize I

$P(I,D)$

I	D	Prob.
i^0	d^0	0.282
i^0	d^1	0.02
i^1	d^0	0.564
i^1	d^1	0.134

$P(D)$

D	Prob.
d^0	0.846
d^1	0.154

Preliminaries: Factors

Factors :

Special case of a "factor", a function, ϕ , which maps input variables to a real value. Conditional probability distribution is a special case of a factor, where the row sum is equal to one.

$P(G \mid I,D)$

	g^1	g^2	g^3
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

Row Sums to one
A probability distribution over "G"

Preliminaries: Factors

Factors :

A Non-probabilistic Example

A	B	ϕ
a^0	b^0	30
a^0	b^1	5
a^1	b^0	1
a^1	b^1	10

Scope: $\{A, B\}$ with value $\{\phi\}$.

Preliminaries: Factors

Factor Product :

$\phi_1(A,B)$

a^1	b^1	0.5
a^1	b^2	0.8
a^2	b^1	0.1
a^2	b^2	0
a^3	b^1	0.3
a^3	b^2	0.9

$\phi_2(B,C)$

b^1	c^1	0.5
b^1	c^2	0.7
b^2	c^1	0.1
b^2	c^2	0.2

$\phi_3(A,B,C)$

a^1	b^1	c^1	$0.5 \cdot 0.5 = 0.25$
a^1	b^1	c^2	$0.5 \cdot 0.7 = 0.35$
a^1	b^2	c^1	$0.8 \cdot 0.1 = 0.08$
a^1	b^2	c^2	$0.8 \cdot 0.2 = 0.16$
a^2	b^1	c^1	$0.1 \cdot 0.5 = 0.05$
a^2	b^1	c^2	$0.1 \cdot 0.7 = 0.07$
a^2	b^2	c^1	$0 \cdot 0.1 = 0$
a^2	b^2	c^2	$0 \cdot 0.2 = 0$
a^3	b^1	c^1	$0.3 \cdot 0.5 = 0.15$
a^3	b^1	c^2	$0.3 \cdot 0.7 = 0.21$
a^3	b^2	c^1	$0.9 \cdot 0.1 = 0.09$
a^3	b^2	c^2	$0.9 \cdot 0.2 = 0.18$

Preliminaries: Factors

Factor Marginalization :

$\phi(A,B,C)$

a^1	b^1	c^1	0.25
a^1	b^1	c^2	0.35
a^1	b^2	c^1	0.08
a^1	b^2	c^2	0.16
a^2	b^1	c^1	0.05
a^2	b^1	c^2	0.07
a^2	b^2	c^1	0
a^2	b^2	c^2	0
a^3	b^1	c^1	0.15
a^3	b^1	c^2	0.21
a^3	b^2	c^1	0.09
a^3	b^2	c^2	0.18

$\phi(A,C)$

a^1	c^1	0.33
a^1	c^2	0.51
a^2	c^1	0.05
a^2	c^2	0.07
a^3	c^1	0.24
a^3	c^2	0.39

Preliminaries: Factors

Factor Reduction :

$$\phi(A,B,C)$$

a^1	b^1	c^1	0.25
a^1	b^1	c^2	0.35
a^1	b^2	c^1	0.08
a^1	b^2	c^2	0.16
a^2	b^1	c^1	0.05
a^2	b^1	c^2	0.07
a^2	b^2	c^1	0
a^2	b^2	c^2	0
a^3	b^1	c^1	0.15
a^3	b^1	c^2	0.21
a^3	b^2	c^1	0.09
a^3	b^2	c^2	0.18

$$\phi(A,B \mid C)$$

a^1	b^1	c^1	0.25
a^1	b^2	c^1	0.08
a^2	b^1	c^1	0.05
a^2	b^2	c^1	0
a^3	b^1	c^1	0.15
a^3	b^2	c^1	0.09

Preliminaries: Factors

Why Factors?

- Fundamental building blocks for distributions in high dimensions.
- Set of basic operations for manipulating these probability distributions.

- We are interested in the joint distribution:

- Exploit conditional independence/ structure. Take a bunch of factors, put them together, order to describe a high-dimensional probability distribution.

Bayesian Network

Random Variables pertaining to a course:

Difficulty
Intelligence
Grade
Letter
SAT

Bayesian Network

Random Variables pertaining to a course:

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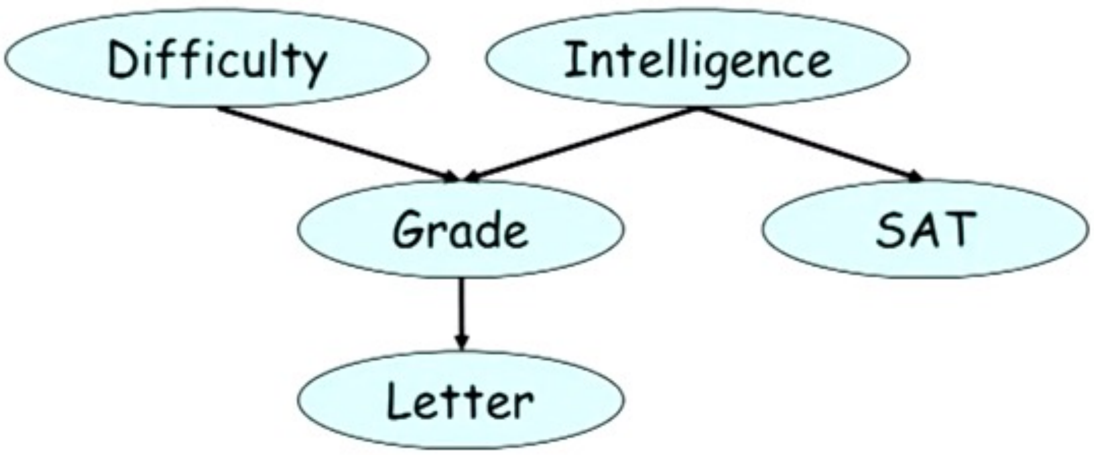
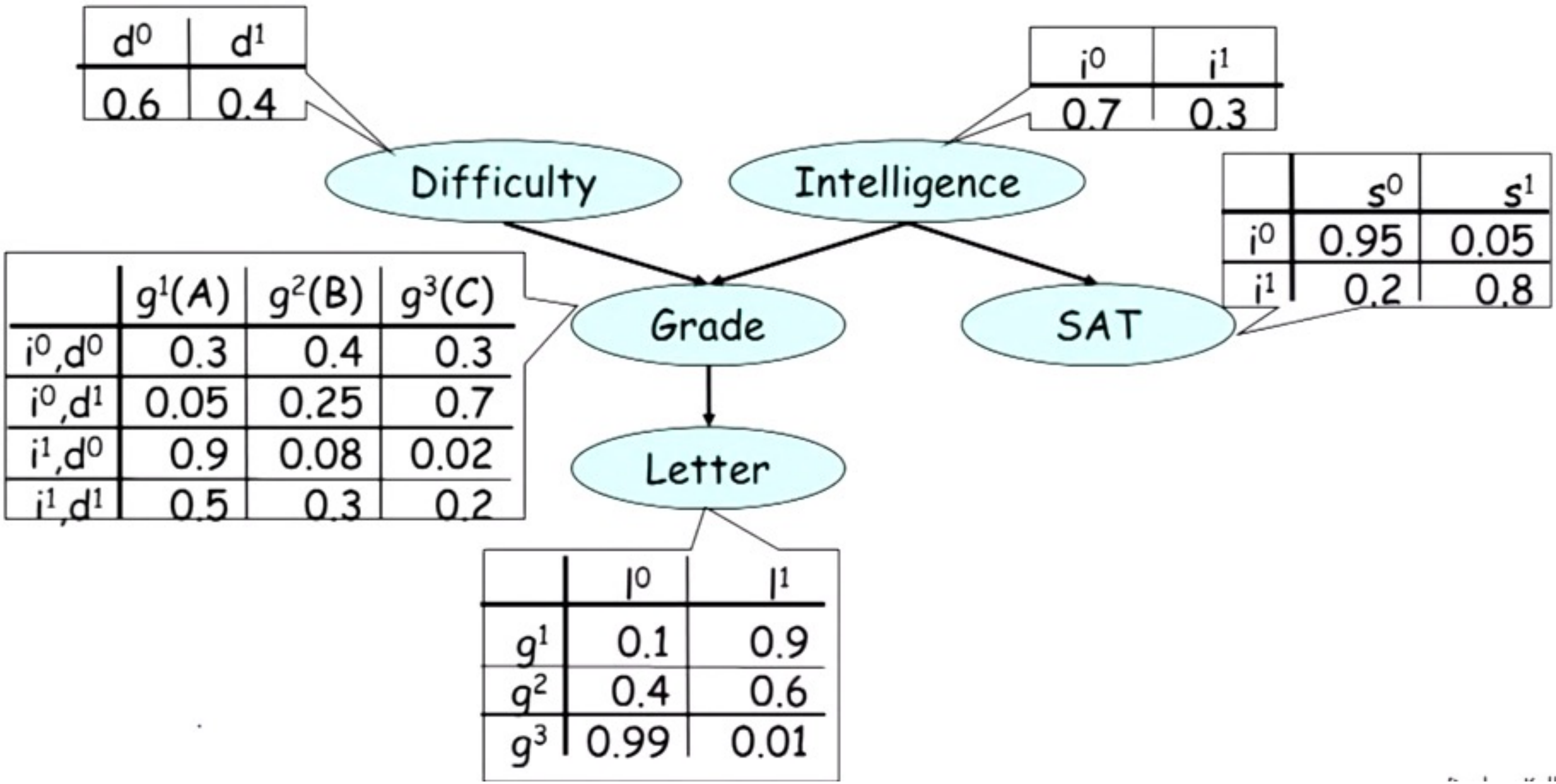
Grade
Letter
SAT



Bayesian Network



d^0	d^1
0.6	0.4

i^0	i^1
0.7	0.3

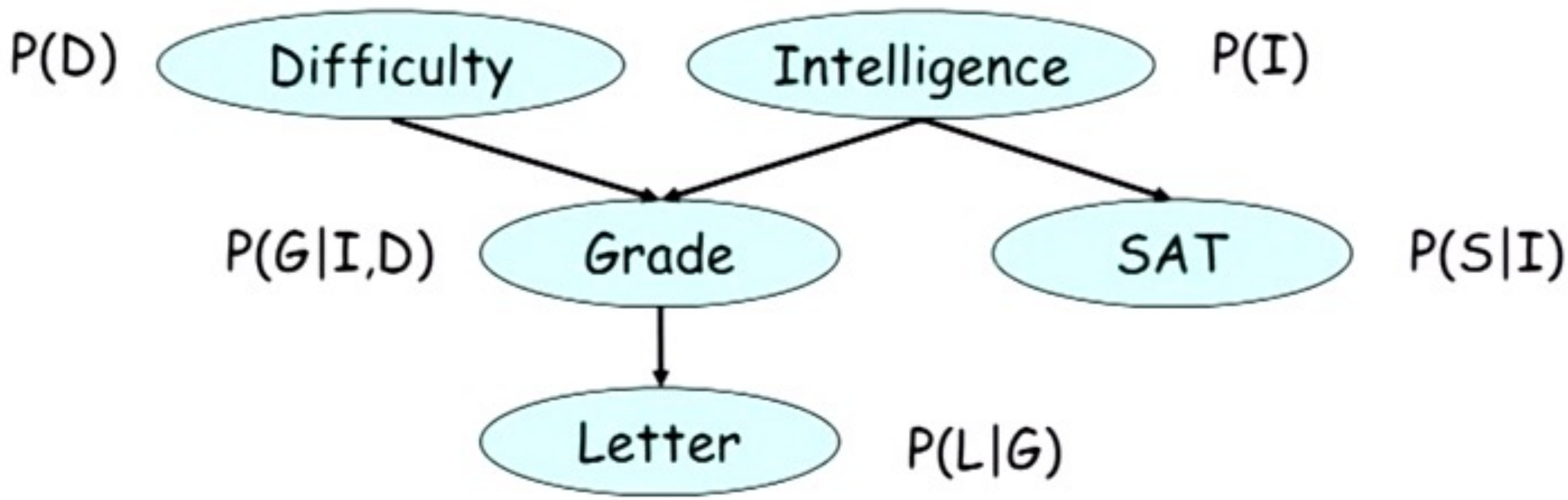
	g^1(A)	g^2(B)	g^3(C)
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^3	0.99	0.01

Fully Parameterized Bayesian Network!

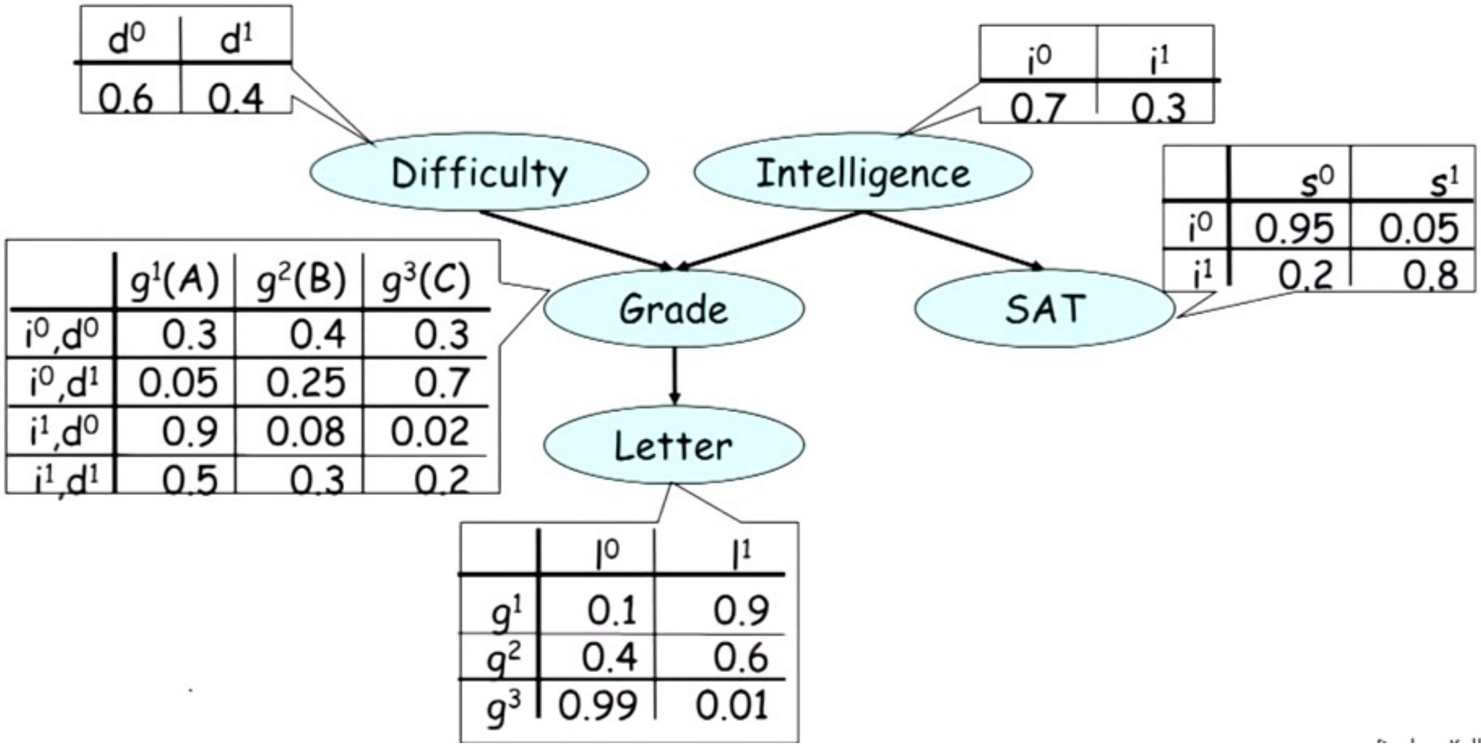
Bayesian Network



$$P(D,I,G,S,L) = P(D)\,P(I)\,P(G|I,D)\,P(S|I)\,P(L|G)$$

This is a factor product!
Big factor (joint distribution) is a product of other factors (conditional distributions)

Bayesian Network



d^0	d^1
0.6	0.4

i^0	i^1
0.7	0.3

	g^1(A)	g^2(B)	g^3(C)
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

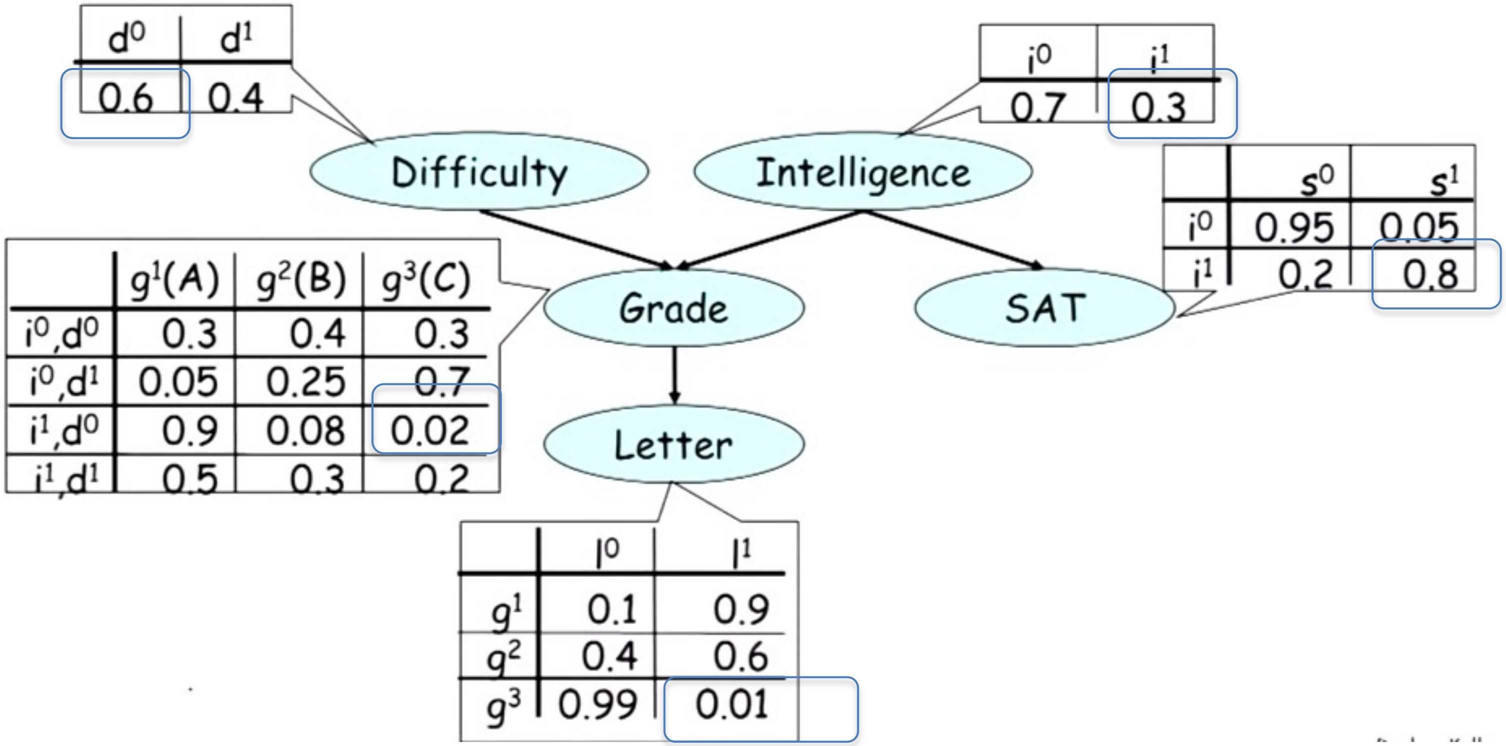
	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^3	0.99	0.01

$$P(D,I,G,S,L) = P(D)P(I)P(G \mid I,D)P(S \mid I)P(L \mid G)$$

Prediction: $P(d^0,i^1,g^3,s^1,l^1) = P(d^0)P(i^1)P(g^3 \mid d^0,i^1)P(s^1 \mid i^1)P(l^1 \mid g^3)$

Fully Parameterized Bayesian Network!

Bayesian Network



d^0	d^1
0.6	0.4

i^0	i^1
0.7	0.3

	$g^1(A)$	$g^2(B)$	$g^3(C)$
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^3	0.99	0.01

$$P(D,I,G,S,L) = P(D)P(I)P(G\mid I,D)P(S\mid I)P(L\mid G)$$

Prediction: $P(d^0,i^1,g^3,s^1,l^1) = P(d^0)P(i^1)P(g^3\mid d^0,i^1)P(s^1\mid i^1)P(l^1\mid g^3)$

$$= 0.6\times 0.03\times 0.02\times 0.8\times 0.01$$

Fully Parameterized Bayesian Network!

Bayesian Network

A Bayesian Network is:

- A directed acyclic graph (DAG) G whose nodes represent the random variables $X_1, X_2, \ldots, X_N$.
- For each node X_i a CPD $P(X_i \mid Par_G(X_i))$.
- The BN represents a joint distribution via the chain rule for Bayesian networks (a factor product):

$$P(X_1, \ldots, X_n) = \Pi_i P\left(X_i \mid Par_G(X_i)\right).$$

Bayesian Network

A Bayesian Network is legal distribution:

1. $P(X_1,\ldots,X_n) \geq 0$ trivial .

2. It sums to one $\sum P(X_1,\ldots,X_n) = 1$ also easy to show

A Bayesian Network factorizes over G:

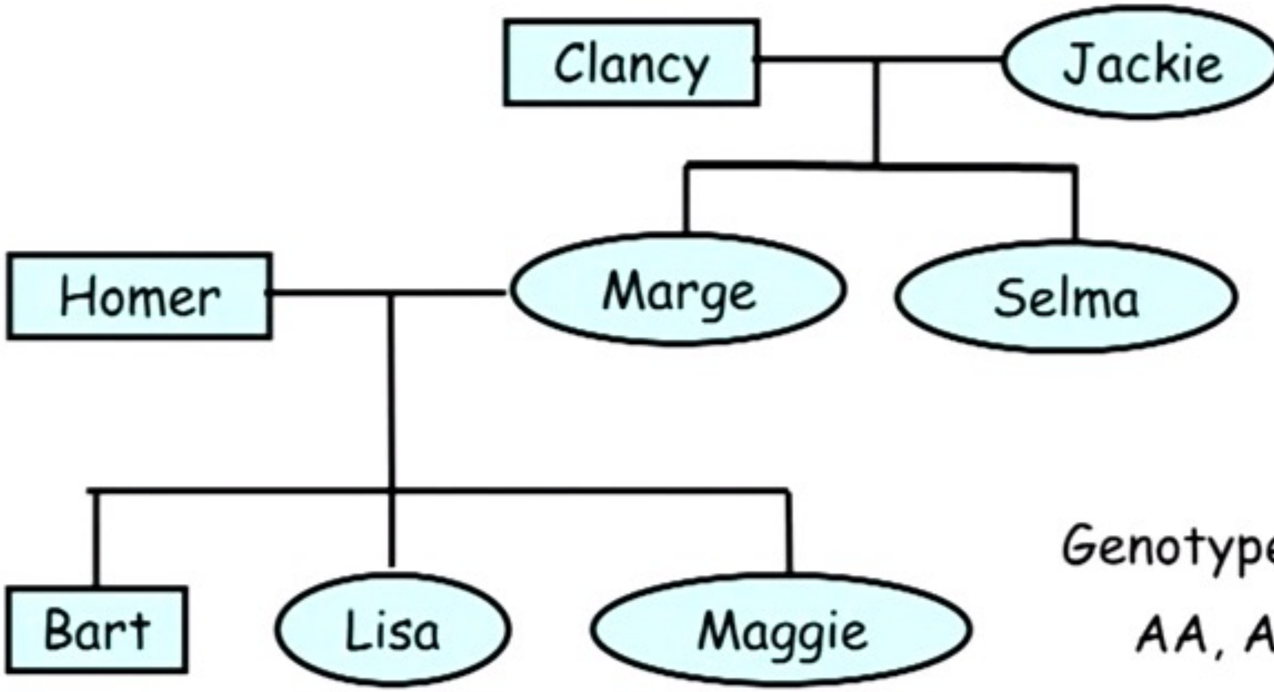
- Let G be a graph over $X_1,\ldots,X_n$.
- P factorizes over G if:

$$P(X_1,\ldots,X_n) = \prod_i P(X_i \mid Par_G(X_i)).$$

Bayesian Network

The first Bayesian Network

Genetic inheritance of blood type



Genotype

AA, AB, AO, BO, BB, OO

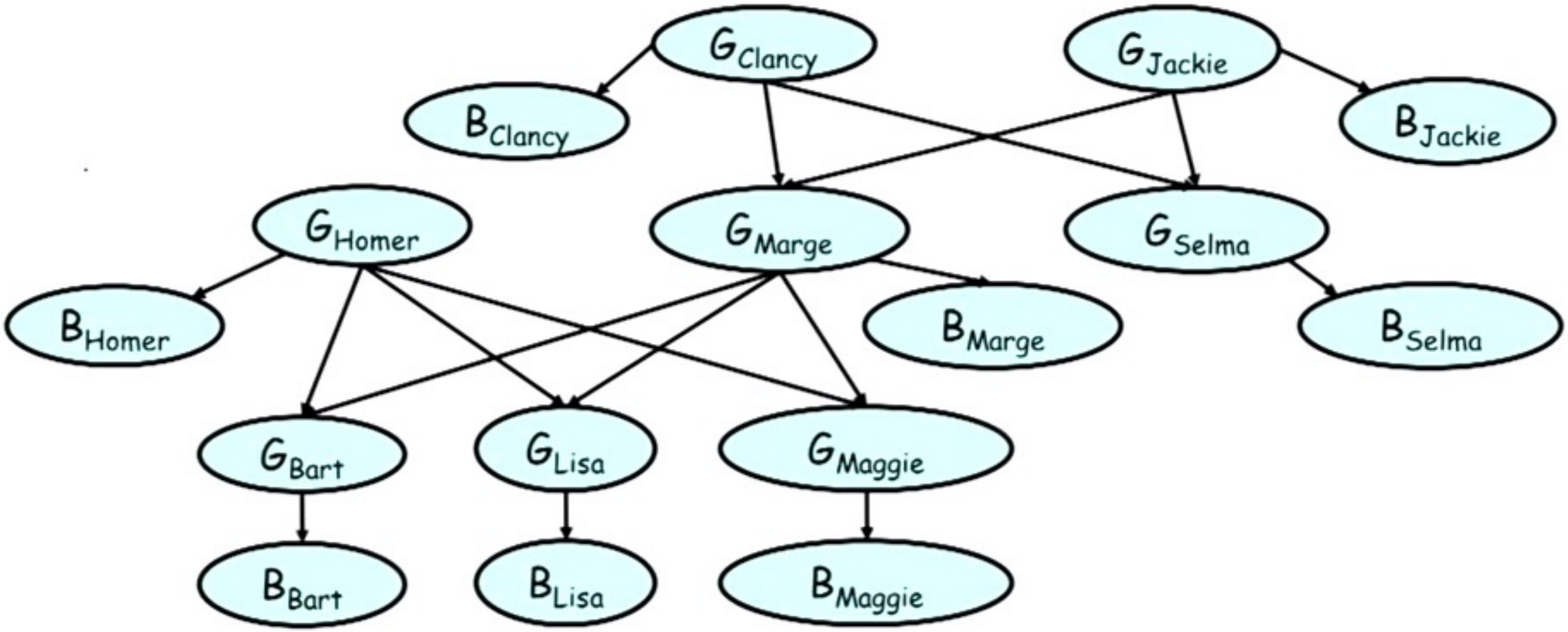
Phenotype

A, B, AB, O

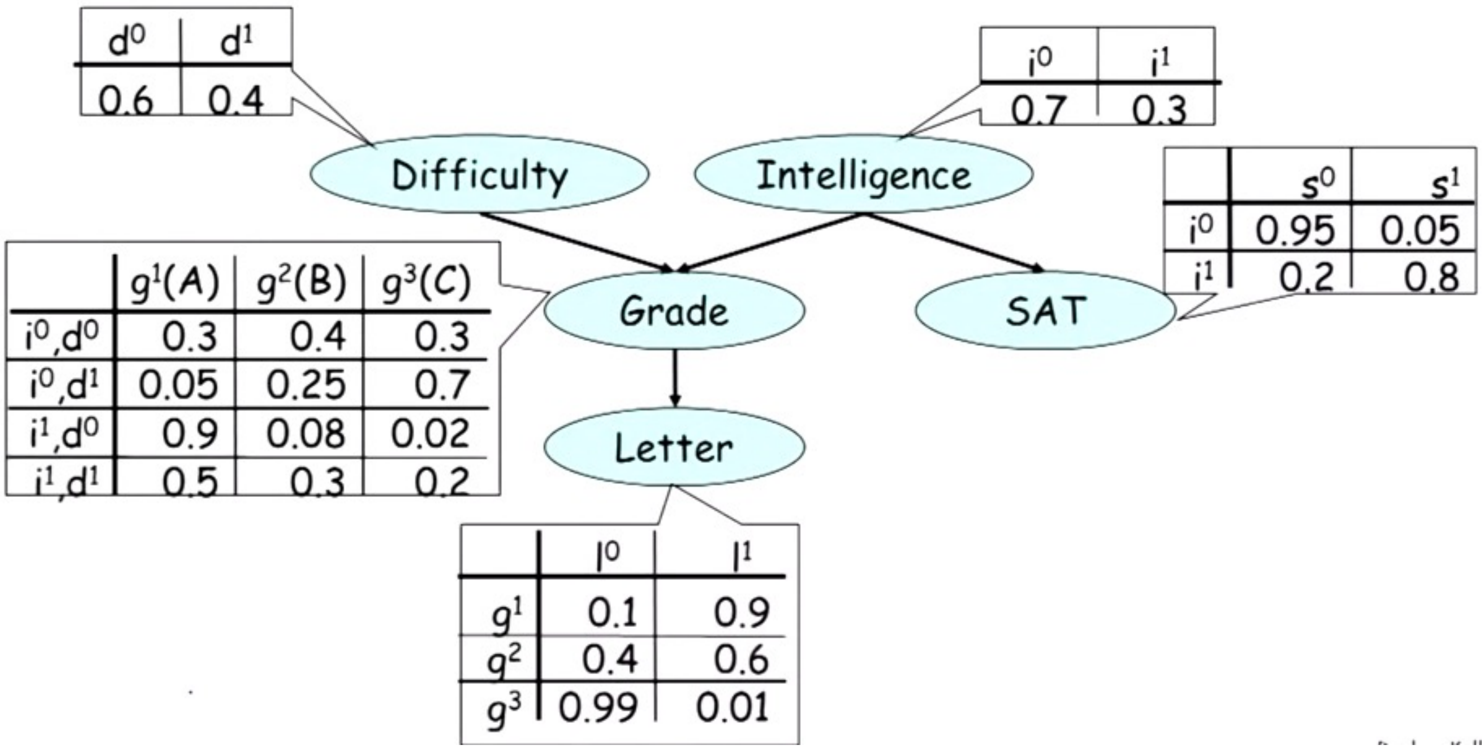
Bayesian Network

Bayesian network for the genetic inheritance of blood type

The purest of Bayesian Networks: full dependence captured.



Reasoning



d^0	d^1
0.6	0.4

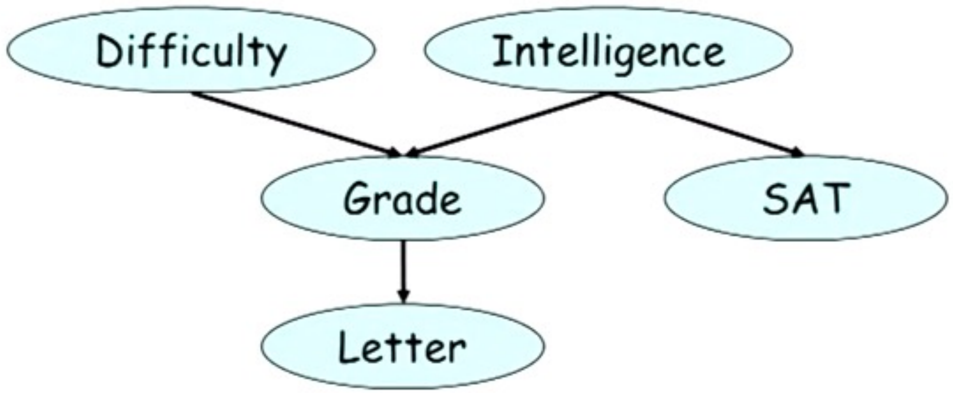
i^0	i^1
0.7	0.3

	$g^1(A)$	$g^2(B)$	$g^3(C)$
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^3	0.99	0.01

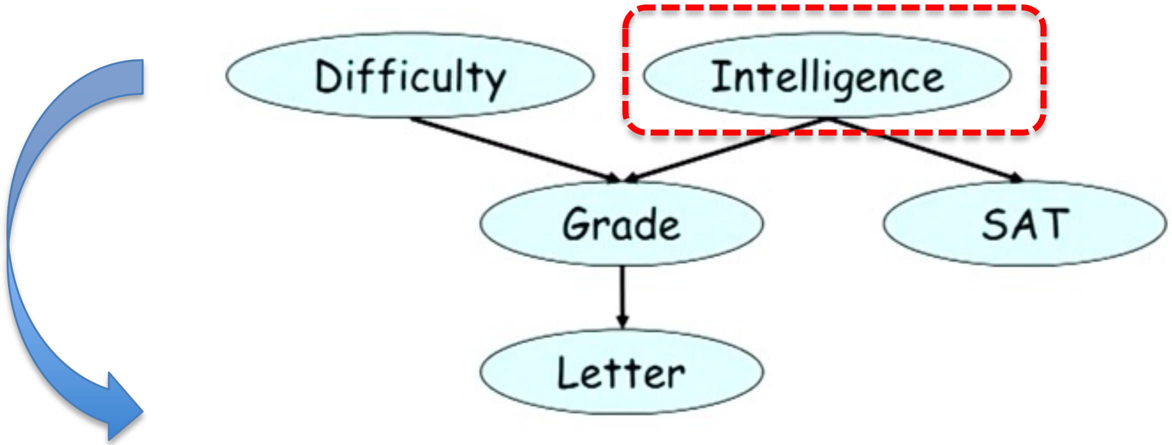
Reasoning



$$P(l^1) = 0.5$$

Reasoning

An Example of **Causal Reasoning:**
Reasoning about the effect(s) given the cause(s).

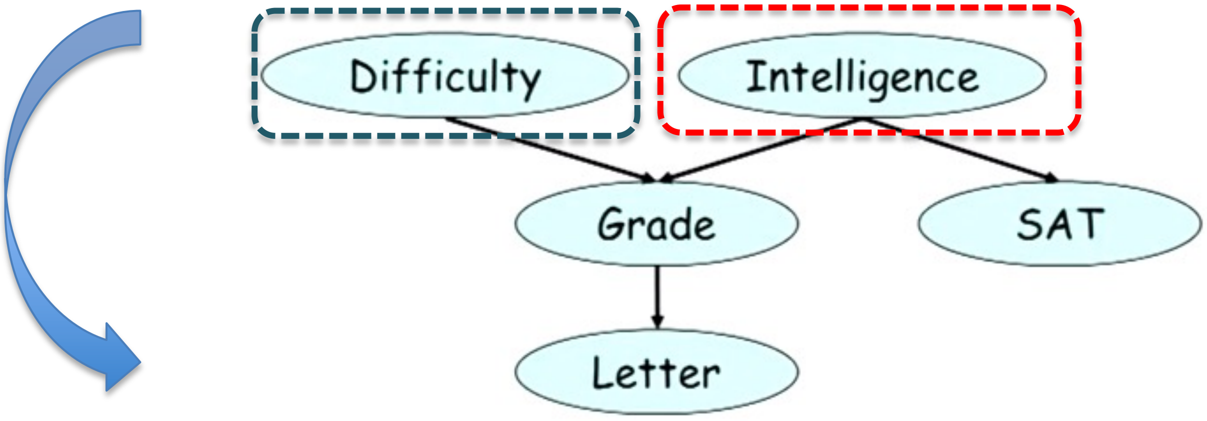


$$P(l^1) = 0.5$$

$$P(l^1 \mid i^0) = 0.39$$

Reasoning

An Example of **Causal Reasoning:**
Reasoning about the effect(s) given the cause(s).



$$P(l^1) = 0.5$$

$$P(l^1 \mid i^0) = 0.39$$

$$P(l^1 \mid i^0, d^0) = 0.51$$

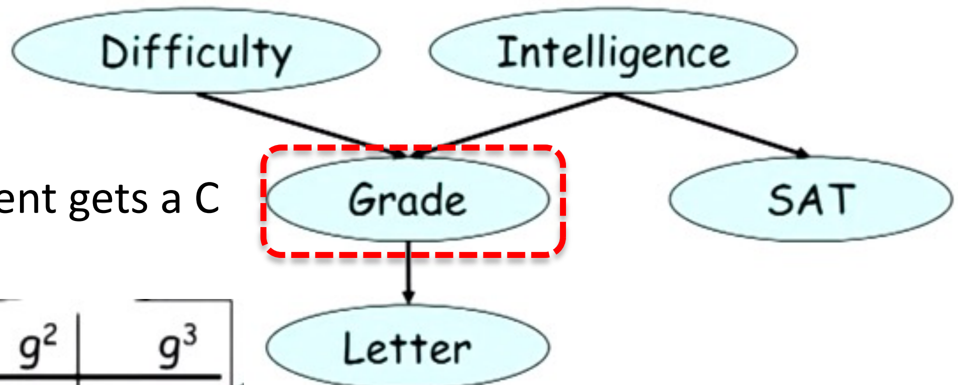
Reasoning

An Example of **Evidential Reasoning:**
Reasoning about the causes given the effect.

$$P(d^1) = 0.4 \qquad P(i^1) = 0.3$$

$$P(d^1 \mid g^3) = 0.39 \qquad P(i^1 \mid g^3) = 0.08$$

Student gets a C



	g^1	g^2	g^3
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

Reasoning

An Example of **Intercausal Reasoning**:
Reasoning in between two causes in light of a single effect.

$$P(d^1) = 0.4$$

$$P(d^1 \mid g^3) = 0.39$$

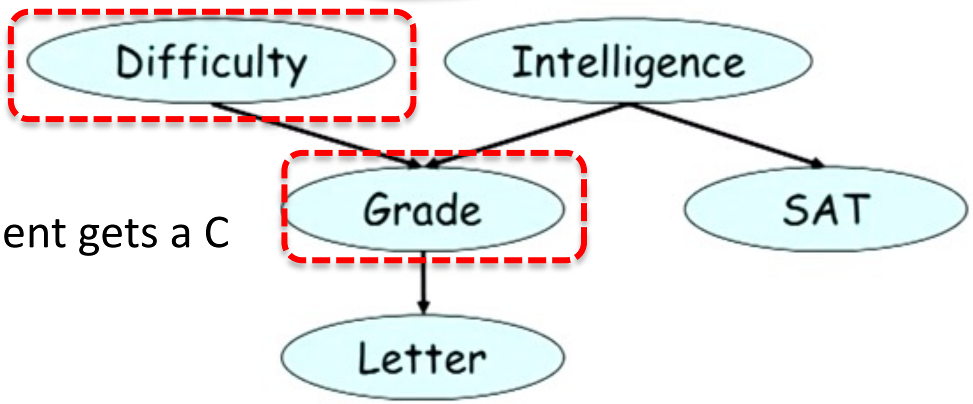
$$P(i^1) = 0.3$$

$$P(i^1 \mid g^3) = 0.08$$

$$P(i^1 \mid g^3, d^1) = 0.11$$

Class is hard

Student gets a C



Reasoning

An Example of **Intercausal Reasoning**:
Reasoning in between two causes in light of a single effect.

$$P(d^1) = 0.4$$

$$P(d^1 \mid g^3) = 0.39$$

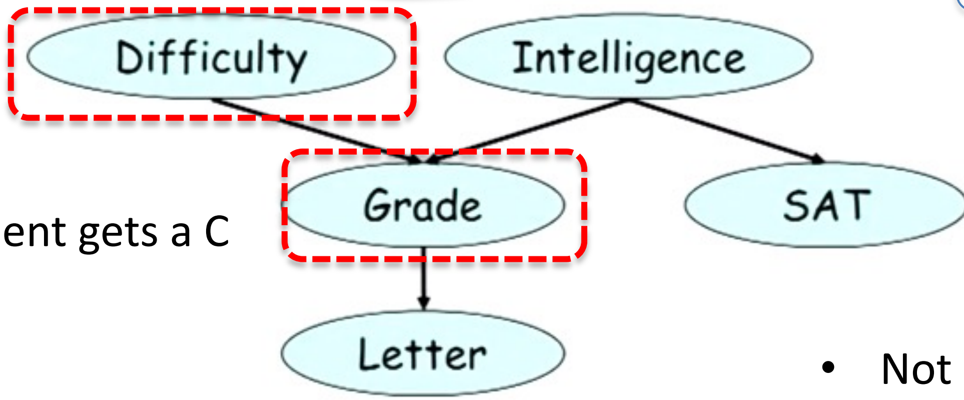
$$P(i^1) = 0.3$$

$$P(i^1 \mid g^3) = 0.08$$

$$P(i^1 \mid g^3, d^1) = \boxed{0.11}$$

Class is hard

Student gets a C



- Not a huge increase, subtle.
- Hard to get a C in this course.
- Accurately reflect the information

In the observational data.

Reasoning

An Example of **Intercausal Reasoning**:
Reasoning in between two causes in light of a single effect.

$$P(i^1) = 0.3$$

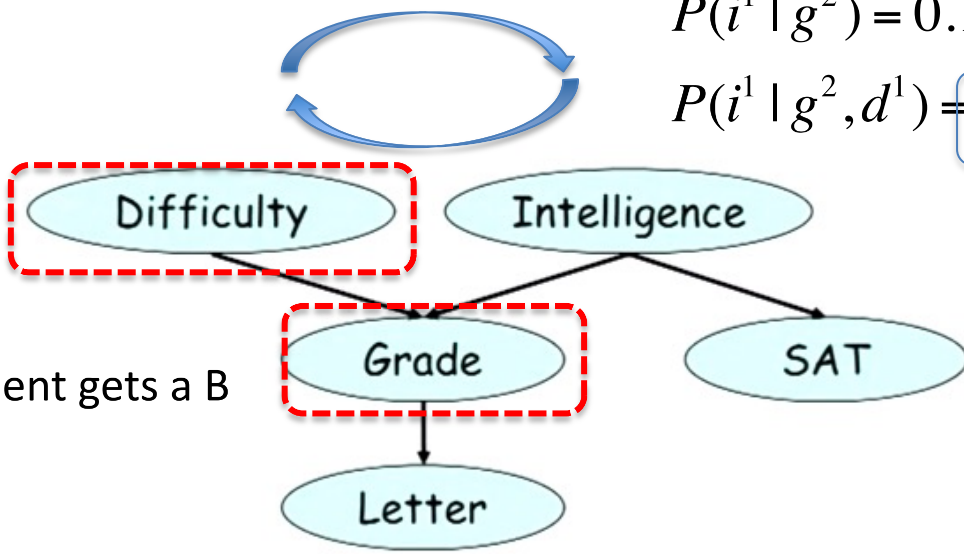
$$P(i^1 \mid g^2) = 0.175$$

$$P(i^1 \mid g^2, d^1) = \boxed{0.34}$$

Big Jump!

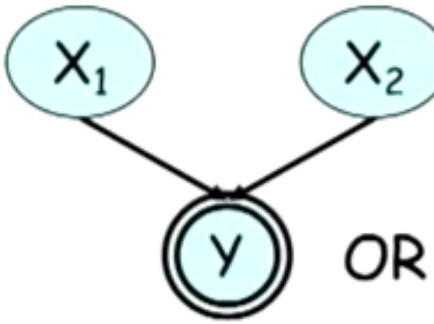
Class is hard

Student gets a B



Reasoning

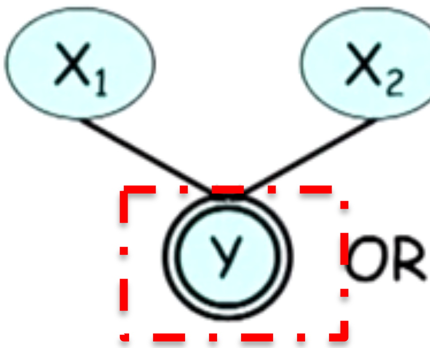
Intercausal Reasoning in Simple Terms:



X_1	X_2	Y	**Prob**
0	0	0	0.25
0	1	1	0.25
1	0	1	0.25
1	1	1	0.25

Reasoning

Intercausal Reasoning in Simple Terms:



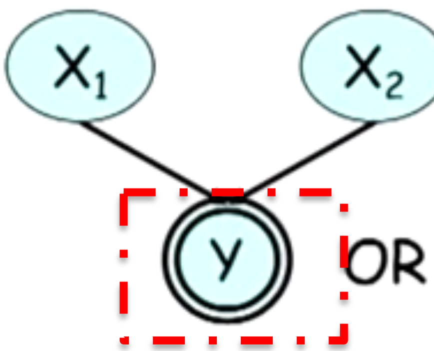
Y=1

X_1	X_2	Y	Prob	
~~0~~	~~0~~	~~0~~	~~0.25~~	
0	1	1	~~0.25~~	0.33
1	0	1	~~0.25~~	0.33
1	1	1	~~0.25~~	0.33

$$P(X_1 = 1) = 2/3$$
$$P(X_2 = 1) = 2/3$$

Reasoning

Intuitively: There are only two ways that Y=1
Either X1=1, or X2 = 1.
If I told you X1 = 1, I have completely **explained away the evidence**.



Y=1

X_1	X_2	Y	Prob	
~~0~~	~~0~~	~~0~~	~~0.25~~	
0	1	1	~~0.25~~	0.33
1	0	1	~~0.25~~	0.33
1	1	1	~~0.25~~	0.33

$$P(X_1 = 1) = 2/3$$

$$P(X_2 = 1) = 2/3$$

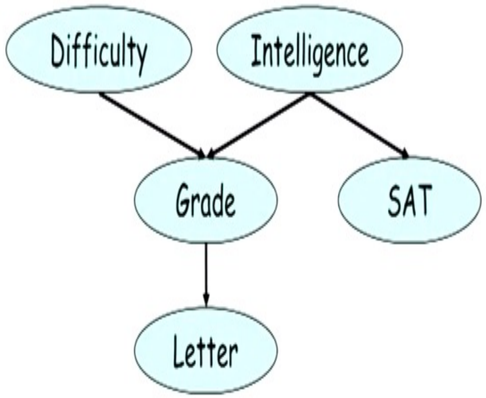
$$P(X_2 = 1 \mid X_1 = 1) = 0.5$$

Bayesian Network

When can X influence Y:

(when we condition on X when does it change our belief in Y).

- X -> Y - yes, straight downward path
- X <- Y – yes, evidential reasoning, belief in Y
- X -> W -> Y – yes, indirect
- X <- W <- Y - yes, reverse
- X <- W -> Y - yes,
- X -> W <- Y - no V-structure/collider model



- A trail X1 –X2 - - Xk, is **active** if it has no v-structures.